

# Notes: Gertler, Kiyotaki & Prestipino (RES, 2020)

## A Macroeconomic Model with Financial Panics

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# 1 Big picture

- **Question.** How can a standard macroeconomic model generate sudden, discrete financial panics, and how do those panics transmit to investment, output, employment, and inflation?
- **Core contribution.** The paper embeds bank rollover crises into a New Keynesian macroeconomic model with investment. This allows a run-like financial collapse to be studied together with output, employment, monetary policy, and nominal rigidities.
- **Main mechanism.** Banks finance partially illiquid long-term assets with short-term liabilities. If creditors refuse to roll over bank debt, banks are forced into liquidation. The liquidation value of bank assets can be below their normal continuation value, making a run self-fulfilling.
- **Balance-sheet complementarity.** Bank runs are possible only in vulnerable states. Weak bank net worth and high leverage make the economy enter a crisis zone. In safe states, the same shock does not generate a run.
- **Nonlinearity.** The model captures the fact that financial crises are not merely large versions of ordinary recessions. A small additional shock can trigger a discontinuous collapse when the economy is close to the run region.
- **Real transmission.** A run lowers bank net worth, raises excess returns and spreads, reduces capital prices, and causes a sharp fall in investment. New Keynesian frictions transmit this financial collapse into output and employment.
- **Quantitative focus.** The model is calibrated to match standard macro moments and financial-intermediation targets, including bank leverage, credit spreads in normal times and crises, and the frequency of systemic banking crises.
- **Crisis experiment.** A sequence of capital-quality shocks can reproduce several key features of the Great Recession: sharp investment decline, financial-sector net-worth collapse, spread spike, and persistent output weakness.
- **Credit-boom extension.** The authors add a belief-driven boom: optimistic beliefs about future capital quality generate credit expansion and make the economy more fragile. The bust occurs when beliefs reverse and the economy enters the run region.
- **Policy implication.** Macroprudential policy is important because individual banks do not internalize how their leverage affects the probability that the entire banking system enters a run equilibrium.

## 2 Model primitives and economic environment

### 2.1 Timeline and aggregate state

The model is an infinite-horizon New Keynesian economy with capital accumulation and financial intermediaries. Time is discrete. The key aggregate state variables include the capital stock, bank net worth, outstanding bank deposits, capital quality, and the realization of the sunspot variable that determines whether a feasible run is actually triggered.

- Production uses capital and labour.
- Capital quality shocks change the effective quantity of productive capital.
- Banks finance capital holdings with net worth and short-term deposits.
- Depositors can decide whether to roll over bank liabilities.
- A sunspot realization selects the run equilibrium when a run is feasible.

**Interpretation:** The model has both fundamental state variables and coordination-driven equilibrium selection. Fundamentals determine whether a run is possible; the sunspot determines whether a possible run happens.

## 2.2 Households

Households contain workers and bankers. Workers supply labour, consume, save through bank deposits, and own firms. Bankers manage intermediaries but eventually exit and return their accumulated wealth to the household.

A representative household has expected lifetime utility of the form

$$E_0 \sum_{t=0}^{\infty} \beta^t u(C_t, L_t), \quad (1)$$

where  $C_t$  is consumption and  $L_t$  is labour supply.

Households choose consumption, labour, and deposits subject to their budget constraint. The relevant household Euler condition is conceptually

$$1 = E_t \left[ \beta \frac{u_C(C_{t+1}, L_{t+1})}{u_C(C_t, L_t)} R_{t+1}^{dep} \right], \quad (2)$$

where  $R_{t+1}^{dep}$  is the gross return on bank deposits.

**Interpretation:** Deposits are the household's safe saving instrument in normal times. Their short-term nature is what makes banks fragile when depositors refuse to roll over.

## 2.3 Capital quality and effective capital

The capital stock used in production is affected by a capital-quality shock. A simple representation is

$$K_t^{eff} = \xi_t S_{t-1}, \quad (3)$$

where  $S_{t-1}$  is the physical capital stock chosen in the previous period and  $\xi_t$  is the capital-quality shock.

**Interpretation:** The shock directly affects the value and productivity of bank assets. Negative capital-quality shocks weaken bank net worth and can move the economy toward the crisis zone.

## 2.4 Banks and the balance sheet

A bank holds capital claims and finances them with net worth and short-term deposits:

$$Q_t S_t^b = N_t + D_t, \quad (4)$$

where  $Q_t$  is the price of capital,  $S_t^b$  is the bank's capital holdings,  $N_t$  is bank net worth, and  $D_t$  is deposits.

A convenient leverage measure is

$$\phi_t = \frac{Q_t S_t^b}{N_t}. \quad (5)$$

**Interpretation:** High leverage increases return on equity in good states but makes the bank more vulnerable to shocks and runs.

## 2.5 Bank net worth dynamics

Bank net worth evolves with the excess return on bank assets relative to liabilities. A reduced-form expression is

$$N_{t+1} = \left[ R_{t+1}^b - R_{t+1}^{dep} \right] Q_t S_t^b + R_{t+1}^{dep} N_t, \quad (6)$$

where  $R_{t+1}^b$  is the return on capital held by banks.

**Interpretation:** Because banks are leveraged, even a modest decline in capital quality or capital price can create a large decline in net worth.

## 2.6 Banker moral hazard and the incentive constraint

Bankers can divert a fraction of assets. Depositors are willing to lend only if continuation value is high enough relative to diversion value. The incentive constraint has the general form

$$V_t^b \geq \lambda Q_t S_t^b, \quad (7)$$

where  $V_t^b$  is the continuation value of the bank and  $\lambda$  measures the diversion technology.

**Interpretation:** The incentive constraint limits leverage in normal times. It is the balance-sheet friction that generates an endogenous spread between bank-intermediated finance and direct household finance.

## 2.7 Runs, insolvency, and illiquidity

A run is a rollover crisis. Depositors refuse to roll over deposits, and the bank must liquidate assets. A run is possible when the liquidation value of bank assets is insufficient to repay short-term creditors.

Let  $Q_t^{liq}$  denote the liquidation price of bank assets. A stylized run condition is

$$Q_t^{liq} S_t^b < D_t. \quad (8)$$

Equivalently, leverage is too high relative to the liquidation value of assets.

**Interpretation:** The model distinguishes illiquidity from fundamental insolvency. A bank can be viable if creditors roll over, but insolvent under fire-sale liquidation if they do not.

## 2.8 The liquidation price

When banks liquidate, households absorb assets directly but face intermediation or management costs. This lowers the asset's liquidation price. The liquidation price is below the normal price because households are less efficient at holding bank-intermediated capital.

**Interpretation:** The liquidation discount is crucial. Without it, a run would merely transfer assets rather than destroy value. With a liquidation discount, runs create real losses.

## 2.9 Production, nominal rigidities, and policy

Final-good producers are monopolistically competitive. Intermediate goods firms produce using capital and labour. Price adjustment costs generate nominal rigidities. Monetary policy follows a Taylor-style rule responding to inflation and output.

A generic monetary-policy rule is

$$R_t = R \left( \frac{\pi_t}{\pi} \right)^{\kappa_\pi} \left( \frac{Y_t}{Y} \right)^{\kappa_y}, \quad (9)$$

where  $\kappa_\pi$  and  $\kappa_y$  are the policy responses to inflation and output.

**Interpretation:** Nominal rigidities matter because financial panics depress demand and investment. Monetary policy and inflation dynamics determine how the shock transmits to real output.

### 3 Model equations and equilibrium logic

#### 3.1 Normal-times equilibrium

In normal times, depositors roll over debt, banks do not default, and bank leverage is constrained by expected continuation values. The relevant equilibrium objects include:

- capital price  $Q_t$ ;
- bank leverage  $\phi_t$ ;
- bank net worth  $N_t$ ;
- deposit return  $R_{t+1}^{dep}$ ;
- return on bank capital  $R_{t+1}^b$ ;
- spreads and excess returns.

**Interpretation:** This equilibrium resembles a standard financial accelerator model: shocks reduce net worth, weaken credit intermediation, and raise spreads.

#### 3.2 Run equilibrium

When the economy is in the crisis zone, two equilibria can exist:

1. **No-run equilibrium:** depositors roll over, banks continue, and assets are valued at normal prices.
2. **Run equilibrium:** depositors refuse to roll over, banks liquidate assets, and capital prices collapse.

**Interpretation:** A run is self-fulfilling because the expectation of liquidation lowers the value of bank assets enough to validate withdrawal.

#### 3.3 Default probability

Let  $\chi$  denote the probability of a sunspot that triggers a run when a run is feasible. The default probability is therefore state dependent:

$$Pr(default_t) = \chi \cdot \mathbf{1}\{\text{run feasible at } t\}. \quad (10)$$

**Interpretation:** The model generates an endogenous jump in default risk. It is not that the sunspot is always powerful; it matters only when balance sheets make a run feasible.

#### 3.4 Excess return and credit spreads

The return spread on bank assets can be written as an excess return relative to a safe rate:

$$X_t = E_t \left[ \Lambda_{t+1} \left( R_{t+1}^b - R_{t+1}^{free} \right) \right]. \quad (11)$$

In the crisis region, the spread rises because bank assets face liquidation risk, default risk, and a collapse in bank net worth.

**Interpretation:** Credit spreads are both an indicator and a transmitter of financial distress. Higher spreads reduce investment demand and depress asset prices further.

### 3.5 Credit boom with optimistic beliefs

The extension introduces a wave of optimism about future capital quality. Bankers come to believe that future capital quality will be high. This raises asset values and supports a credit boom.

A stylized belief process is

$$E_t^b[\xi_{t+h}] > E_t[\xi_{t+h}], \quad (12)$$

where  $E_t^b$  denotes bankers' subjective expectation.

**Interpretation:** The boom is not caused by an actual improvement in fundamentals. It is caused by optimistic beliefs that increase leverage and debt accumulation, making the system more fragile.

### 3.6 Belief reversal and crisis

When optimism fades, the economy has higher debt and bank leverage than it otherwise would. This makes it easier for modest adverse shocks to place the economy in the crisis zone.

**Interpretation:** The belief shock does not need to directly cause the panic. It creates the fragile conditions under which a later shock can generate a panic.

## 4 Calibration and numerical design

### 4.1 Standard macro parameters

The standard parameters include the discount factor, risk aversion, Frisch elasticity, elasticity of substitution, capital share, depreciation, investment adjustment cost parameters, and government spending share.

- The discount factor targets the risk-free rate.
- The capital share targets the observed capital share in output.
- The depreciation rate is set to match quarterly investment relative to capital.
- Price adjustment costs target an average markup and a Phillips curve slope.
- Monetary-policy parameters follow the Taylor-rule literature.

**Interpretation:** The macro block is deliberately conventional. The paper wants the nonlinearity to come from banking panics rather than from unusual macro assumptions.

### 4.2 Financial-intermediation parameters

The financial-sector parameters include:

- banker survival probability;
- initial banker net-worth endowment;
- share of diverted assets;
- threshold for household direct-finance costs;
- marginal household intermediation cost;
- sunspot probability;
- volatility and persistence of capital-quality shocks.

**Interpretation:** These parameters govern leverage, spreads, bank net worth, and the feasibility of runs.

### 4.3 Calibration targets

The paper uses targets such as:

- normal-times bank leverage;
- investment drop during crisis episodes;
- crisis-level credit spreads;
- frequency of systemic banking crises;
- unconditional volatility of investment and macro aggregates.

**Interpretation:** The model is not estimated. It is calibrated to discipline the key macro-financial margins relevant for crisis dynamics.

### 4.4 Numerical experiments

The paper studies several experiments:

1. a capital-quality shock without a run;
2. a sequence of shocks that moves the economy into the run region;
3. a comparison of New Keynesian and flex-price economies;
4. a Great Recession-style model-versus-data crisis simulation;
5. a belief-driven credit boom followed by a bust.

**Interpretation:** The experiments separate three mechanisms: balance-sheet amplification, rollover panic, and belief-driven fragility.

## 5 Identification, interpretation, and what the model explains

### 5.1 What the model is designed to explain

The model is designed to explain the following crisis facts:

- financial panics are sudden and discontinuous;
- crises can occur without a large exogenous fundamental shock;
- credit spreads can spike sharply;
- financial-sector net worth can collapse;
- investment falls more sharply than consumption;
- credit booms often precede banking crises.

**Interpretation:** The model is an explanation of crisis nonlinearities, not just average business-cycle dynamics.



## 5.2 Role of nominal rigidities

A flex-price version of the model generates a smaller output collapse. Nominal rigidities increase the real impact of a financial panic because the collapse in investment demand transmits to output and employment rather than being offset by flexible price adjustment.

**Interpretation:** The financial crisis is amplified by the New Keynesian block. Financial panics and nominal rigidities are complements.

## 5.3 Why credit booms increase fragility

Optimistic beliefs raise investment and debt before the crisis. The economy moves closer to the run region because bank balance sheets become more exposed. When optimism ends, the debt overhang and weak bank net worth amplify the bust.

**Interpretation:** The model explains why booms can be dangerous even if they do not immediately produce visible stress.

## 5.4 Macroprudential logic

Individual banks do not internalize how their leverage affects the probability of a system-wide run. Because run probability depends on aggregate balance sheets, leverage choices create an externality.

**Interpretation:** This externality gives a role for macroprudential policy: limiting leverage in booms may reduce the probability of a discontinuous collapse.

## 5.5 Model limitations

- The run is selected by a sunspot when the run region exists.
- The model abstracts from many institutional details of shadow banking.
- The zero lower bound is not imposed in the numerical experiments.
- Belief dynamics are stylized rather than derived from learning or information frictions.
- The calibration is designed to match crisis moments, not to estimate all structural parameters.

**Interpretation:** The model's strength is the clean integration of bank runs with a macro model. Its weakness is that several crisis triggers and institutional details are intentionally reduced-form.

## 6 Tables and figures

### 6.1 Table 1 and Table 2: Calibration and model moments

Read:

- Table 1 reports the standard macro parameters and the financial-intermediation parameters.
- The financial parameters target leverage, crisis spreads, crisis frequency, and investment volatility.
- Table 2 compares unconditional standard deviations in the model with U.S. data from 1983Q1 to 2007Q3.
- The model matches the volatility of investment reasonably well but makes consumption more volatile than in the data.

**Economic message:** The calibration makes the model conventional on the macro side and disciplined on the financial-crisis side. This gives credibility to the experiments that follow.

| Parameter                                  | Description                                         | Value  | Target                             |
|--------------------------------------------|-----------------------------------------------------|--------|------------------------------------|
| <b>Standard parameters</b>                 |                                                     |        |                                    |
| $\beta$                                    | Impatience                                          | 0.99   | Risk Free Rate                     |
| $\gamma_h$                                 | Risk Aversion                                       | 2      | Literature                         |
| $\varphi$                                  | Inverse Frisch Elasticity                           | 0.5    | Literature                         |
| $\epsilon$                                 | Elasticity of Subst. Across Varieties               | 11     | Markup 10%                         |
| $\alpha$                                   | Capital Share                                       | 0.33   | Capital Share                      |
| $\delta$                                   | Capital Depreciation                                | 0.33   | $\frac{I}{K} = 0.025$              |
| $\eta$                                     | Elasticity of Q w.r.t. I                            | 0.25   | Literature                         |
| $a$                                        | Inv. Technology                                     | 0.53   | Q=1                                |
| $b$                                        | Inv. Technology                                     | -0.83% | $\Gamma(I) = I$                    |
| $G$                                        | Government Expenditure                              | 0.45   | $\frac{G}{Y} = 0.2$                |
| $\rho^p$                                   | Price Adj. Costs                                    | 1000   | Slope of Phillips curve = 0.01     |
| $\kappa_\pi$                               | Policy Response to Inflation                        | 1.5    | Literature                         |
| $\kappa_y$                                 | Policy Response to Output                           | 0.125  | Literature                         |
| <b>Financial intermediation parameters</b> |                                                     |        |                                    |
| $\sigma$                                   | Banker Survival Rate                                | 0.93   | Leverage = 10                      |
| $\zeta$                                    | New Banker Endowment                                | 0.1%   | Investment Drop in Crisis = 35%    |
| $\theta$                                   | Share of Divertible Assets                          | 0.22   | Spread Increase in Crisis = 1.5%   |
| $\gamma$                                   | Threshold for HH Intermediation Costs               | 0.61   | $\frac{s^b}{s} = 0.33$             |
| $\chi$                                     | Marginal HH Intermediation Costs                    | 0.105  | $ER^b - R = 2\%$ Annual            |
| $\kappa$                                   | Sunspot Probability                                 | 0.15   | Run Probability = 4% Annual        |
| $\sigma(\epsilon^k)$                       | Standard Deviation of Innovation to Capital Quality | 0.5%   | Standard Deviation of Output (C+I) |
| $\rho^k$                                   | Serial Correlation of Capital Quality               | 0.7    | Standard Deviation of Investment   |

(a) Table 1: parameter choices.

|     | Data<br>1983-2007 Q3 | Model<br>No runs happen |
|-----|----------------------|-------------------------|
| Y   | 1.9                  | 1.4                     |
| C+I | 2.7                  | 2.9                     |
| I   | 7.2                  | 7.1                     |
| C   | 1.3                  | 2                       |
| L   | 3.1                  | 2.6                     |

Notes: For output, investment, consumption, and government expenditure, we compute real per capita values by dividing by population and adjusting by the GDP deflator. For labour, we compute per capita hours worked by dividing total hours by population. We then report standard deviations of the logged variables in deviations from a linear trend starting in 1983 Q1 and ending in 2007 Q3.  
Source: Output, investment (gross domestic investment plus durable good consumption), consumption (personal consumption expenditure less durables), government expenditure, and GDP deflator are from the Bureau of Economic Analysis. Total labour hours (aggregate hours, non-farm payrolls) and population (civilian non-institutional, 16 years and

(b) Table 2: data versus model moments.

### 6.2 Figure 1 and Figure 2: Capital-quality shocks and run versus no-run dynamics

Read:

- Figure 1 shows a negative capital-quality shock that does not generate a run.
- Capital quality falls, bank net worth declines, leverage rises, and investment/output fall, but the economy stays outside the run region.
- Figure 2 uses a sequence of shocks that pushes the economy into the run region.
- When the run equilibrium is selected, bank net worth collapses, leverage spikes, spreads jump, and investment falls much more sharply than in the no-run counterfactual.

**Economic message:** Balance-sheet weakness is not enough by itself to guarantee a panic. But once the economy is close to the crisis zone, a small additional shock can trigger a discontinuous collapse.

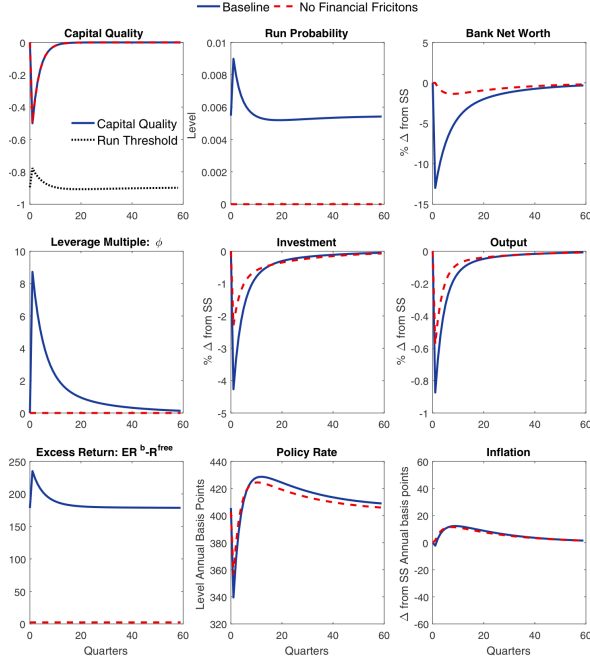


Figure 1: no-run response to a capital-quality shock

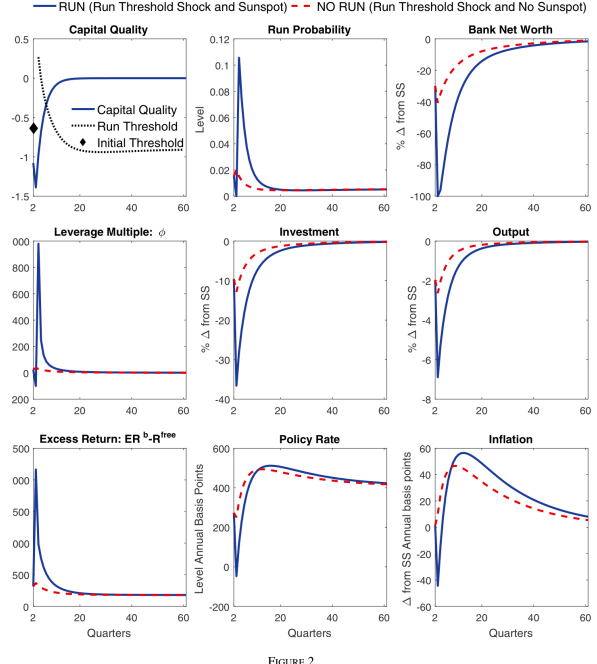


Figure 2: run versus no-run response

### 6.3 Figure 3 and Figure 4: Flex-price comparison and nominal rigidities

Read:

- Figure 3 shows the same shock sequence in a flex-price economy.
- The flex-price model still generates financial amplification, but the real contraction is smaller and less persistent than in the baseline New Keynesian economy.
- Figure 4 decomposes the role of nominal rigidities by comparing capital prices, dividend values, real-rate components, spreads, run probability, and employment.
- Nominal rigidities affect the price of capital and the macro response through real rates, spreads, and employment adjustment.

**Economic message:** The banking panic is a financial mechanism, but nominal rigidities determine how strongly it moves real activity. Financial frictions and New Keynesian frictions amplify each other.

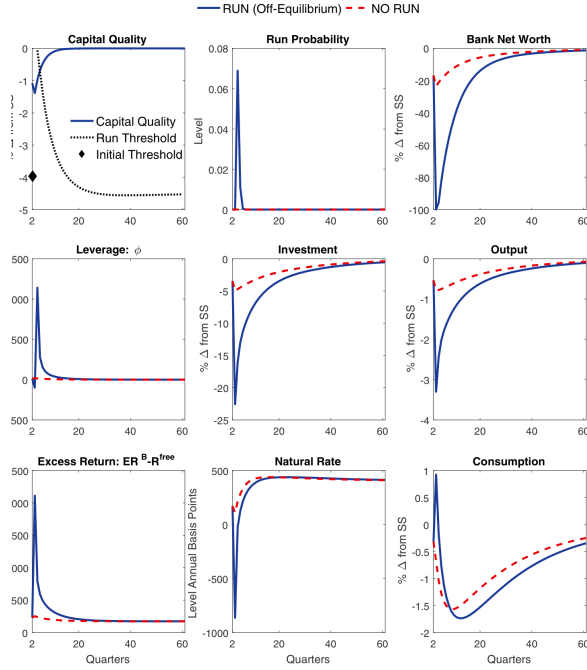


Figure 3: flex-price run versus no-run

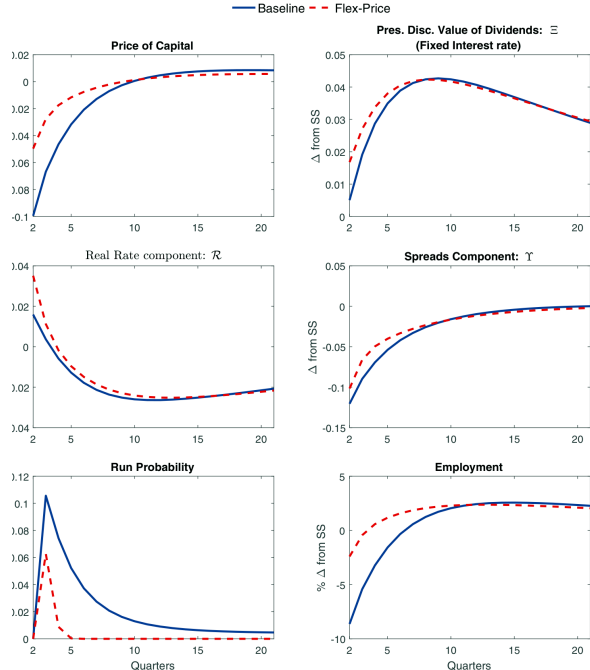


Figure 4: role of nominal rigidities

## 6.4 Figure 5: Model versus data in the financial crisis

Read:

- Figure 5 compares the model to data for investment, the XLF index and net worth, credit spreads, GDP, labour hours, and consumption.
- The blue line is data, the red dashed line is the run model, and the green dotted line is the no-run counterfactual.
- The run model captures the sharp fall in investment and financial net worth and the spike in spreads.
- The no-run counterfactual produces a much milder contraction.

**Economic message:** The model's main quantitative point is that a run-like financial panic is needed to replicate the amplitude of the Great Recession collapse in investment and financial-sector balance sheets.

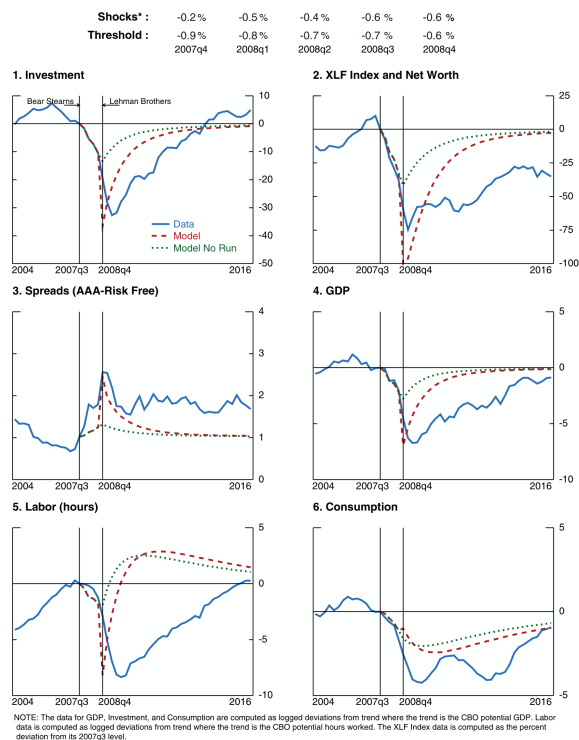


FIGURE 5  
Banking panics: the role of nominal rigidities

## 6.5 Figure 6 and Figure 7: Credit boom, beliefs, and crisis after the boom

Read:

- Figure 6 shows the belief-driven boom: optimism about future capital quality raises debt and increases the probability of entering the crisis zone.
- The economy becomes fragile before the crisis, even without a large realized adverse fundamental shock.
- Figure 7 repeats the crisis simulation after the credit boom.
- The model again produces a collapse in investment, financial-sector net worth, GDP, labour, and consumption similar to the data.

**Economic message:** Credit booms matter because they increase fragility. Optimism raises leverage and debt, making the economy more vulnerable to a panic when beliefs reverse.

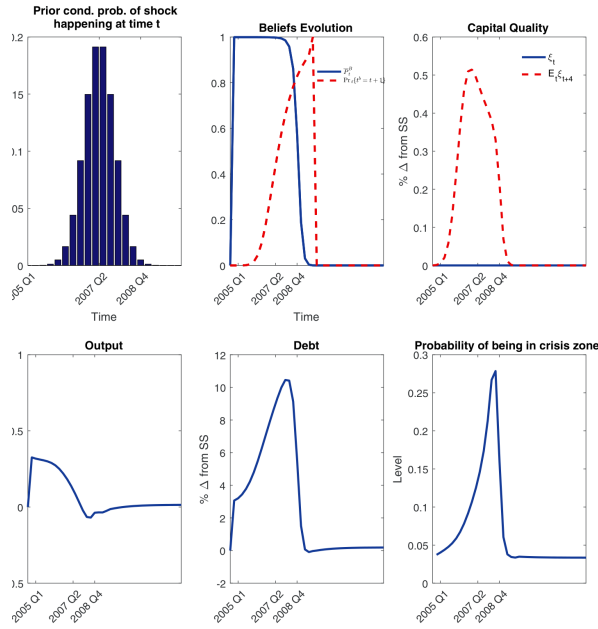


FIGURE 6  
Belief dynamics and credit boom

Figure 6: belief dynamics and credit boom

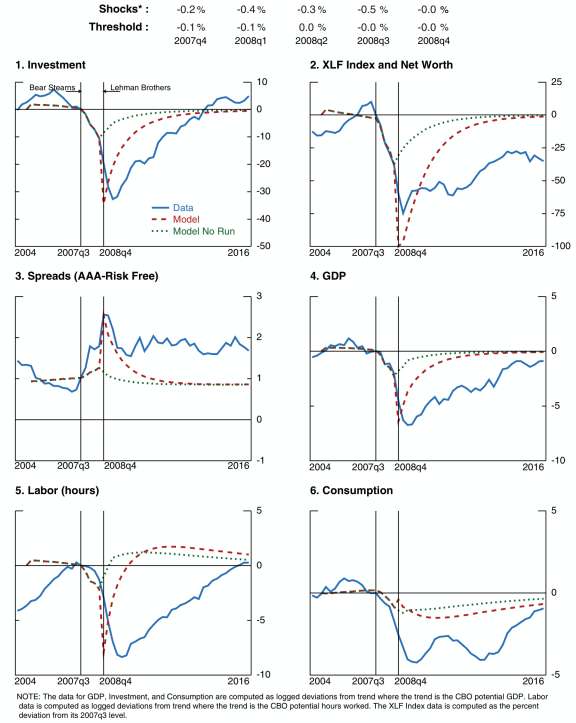


FIGURE 7  
Financial crisis after credit boom: model versus data

Figure 7: crisis after credit boom

## 6.6 Figure 8 and Figure A1: Forecast errors and the run threshold

Read:

- Figure 8 compares model-generated forecast errors in excess returns with survey evidence on credit-spread forecast errors.
- The model generates optimism before the crisis and partial pessimism afterward, broadly consistent with the survey evidence.
- Figure A1 illustrates the threshold capital-quality logic in the appendix.
- The figure shows how leverage determines the threshold below which a run becomes feasible.

**Economic message:** The model does not only match crisis quantities; it also produces belief errors that resemble observed credit-spread forecast mistakes. The appendix figure clarifies the leverage-threshold mechanism behind the crisis zone.

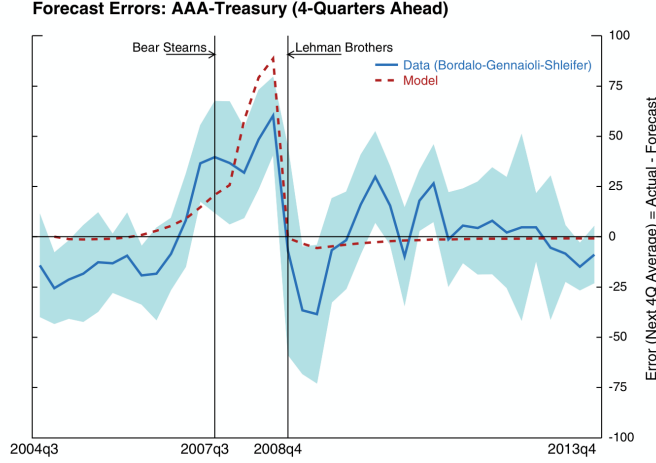


FIGURE 8  
Forecast errors in excess returns: model versus data

(a) Figure 8: forecast errors.

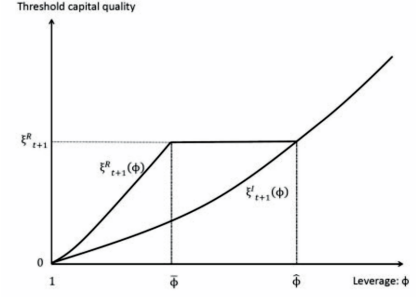


FIGURE A1

(b) Figure A1: threshold capital quality.

## 7 Short summary

- The paper builds a New Keynesian model with banks, investment, and self-fulfilling rollover crises.
- Banks hold long-term illiquid assets financed by short-term deposits.
- A run is possible when the liquidation value of bank assets is below deposit liabilities.
- Runs are state dependent: they occur only when balance sheets place the economy in a crisis zone.
- A sunspot selects the run equilibrium once the run equilibrium exists.
- Runs produce large drops in bank net worth, investment, output, and employment.
- Nominal rigidities amplify the real effects of the financial panic.
- The model can reproduce key features of the Great Recession when a run occurs in 2008Q4.
- A belief-driven credit boom increases leverage and debt before the crisis, making the economy fragile.
- The paper provides a macroprudential rationale: individual banks do not internalize how leverage raises system-wide run risk.

## 8 Conclusion

### 8.1 Core conclusion

Gertler, Kiyotaki, and Prestipino develop a macroeconomic model in which a banking panic can arise endogenously as a self-fulfilling rollover crisis. The key insight is that bank runs are possible only in vulnerable balance-sheet states, but when they occur they produce a sudden and severe collapse in credit intermediation and real activity.

### 8.2 Economic intuition

Banks normally create value by intermediating assets that households cannot efficiently hold directly. But banks finance those assets with short-term debt. If creditors refuse to roll over, assets must be liquidated at a

discount. When leverage is high enough, the expectation of liquidation makes withdrawal rational, creating a self-fulfilling run. Because investment depends on bank net worth and capital prices, the run spills into the real economy.

### **8.3 Takeaway**

The paper shows why macroeconomic models of crises need both balance-sheet fragility and coordination failure. Balance sheets determine vulnerability; sunspot-driven rollover failure creates the discontinuity. The model also shows why credit booms can be dangerous and why macroprudential policy is needed before the panic occurs.